

Mohammed Aljomaily

Portfolio

Computer Engineering Student | Embedded Systems, PCB Design & RF

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Preface

My name is Mohammed Aljomaily. I am a Computer Engineering student at Istanbul Aydın University, and I am applying for an internship at Baykar.

I came into engineering through software, somewhat by accident. During a gap year I took an internship at Applab in Doha, and it turned into fourteen months of real work: I built backend services for an AI video platform used by Al Jazeera, and later an AI chatbot for the General Authority of Customs. It was serious work and I am proud of it. But it was never the part of engineering I went looking for. The thing I actually did with my own time was different. I was designing circuit boards, etching them at my kitchen table, soldering them, and flying what I had built.

That is the work I want to do properly.

Over the past year I have designed and hand-etched my own PCBs at home, because paying for fabrication and shipping to Turkey did not make sense for the size of the boards I was making. I designed a two-layer ESP32 board carrying two nRF24L01 transceivers on separate SPI buses, and discovered after the first layout that two layers are close to impossible to produce by hand. I widened the vias, reworked the routing, and made it again. The board works. I have built and tuned a 5-inch FPV quadcopter and written the firmware for the radio link on my fixed-wing designs. I published a KiCad footprint library for the SSD1306 OLED because I needed one that worked in a current version of KiCad and could not find it, and I use it in my own boards. I keep a written preflight checklist for my aircraft, because I would rather catch a loose propeller on the bench than in the air.

Baykar is one of a small number of places where the thing I do for my own interest is the thing the company does at scale. I want to take problems and make real-world solutions that combine both hardware and software.

I am not arriving with only coursework behind me. I can take a design from a schematic to a board I can hold in my hand, and I have done that several times. I work in C and C++ on ESP32 and ESP8266, in KiCad for boards and FreeCAD for mechanical parts, and I am comfortable at a bench with a soldering iron and a multimeter. I also know how to write software that keeps running after the demo, which I learned the hard way on systems that real people depended on.

The rest of this document is a record of what I have designed, built and flown. Almost everything here was made at a desk at home, including the circuit boards themselves. Where a project is unfinished I have said so.

5-Inch FPV Quadcopter

2026

A complete 5-inch FPV aircraft, built from parts and configured from scratch: **Mario 5** frame, **Mamba MK4 F722** flight controller, **F40 V2** motors, and **DJI O4 Lite** video on **6S**. I configured Betaflight, set up the **ExpressLRS 2.4 GHz** control link between a RadioMaster Pocket running EdgeTX and the receiver, and tuned the aircraft to fly the way I wanted it to rather than the way it arrived.

I wrote the preflight checklist for this aircraft myself, as a ground-check procedure I run before every session. It is not a formality: on a machine with propellers, batteries and a radio link, the failures that hurt are the boring ones.

Alongside the airframe I built out the rest of the workflow, including GyroFlow stabilisation for footage from a GoPro HERO7 with ND filters, and a custom EdgeTX sound pack and 3D-printed stick ends for the radio.



Figure 1: The finished 5-inch build.

Dual-Transceiver ESP32 Jammer Board

2026

A two-layer ESP32 jammer board I designed in **KiCad** for educational study of 2.4 GHz protocols, and then fabricated by hand at home: an **ESP32 DevKit V1** carrying **two nRF24L01+PA+LNA** transceivers, each on its own SPI bus (HSPI and VSPI) so both radios operate independently, plus an SSD1306 OLED for status and an optional TP4056 charging circuit for a Li-Ion cell. The finished board is 64.14×77.47 mm.

I did not use a fabrication house. Shipping to Turkey plus the cost of a local service made no sense for boards this size, so I generate the artwork, transfer it, etch it, drill it and assemble it myself. That decision shaped the design: the first layout was two layers and turned out to be close to impossible to produce by hand, so I enlarged the vias, reworked the routing for the process, cleared every DRC error, and made it a second time. The second board works.

The firmware is adapted from an existing open-source project rather than written from scratch. I wanted my hardware to be fully compatible with it, so the engineering work on my side was the board, the schematic capture, the custom footprints and the fabrication.

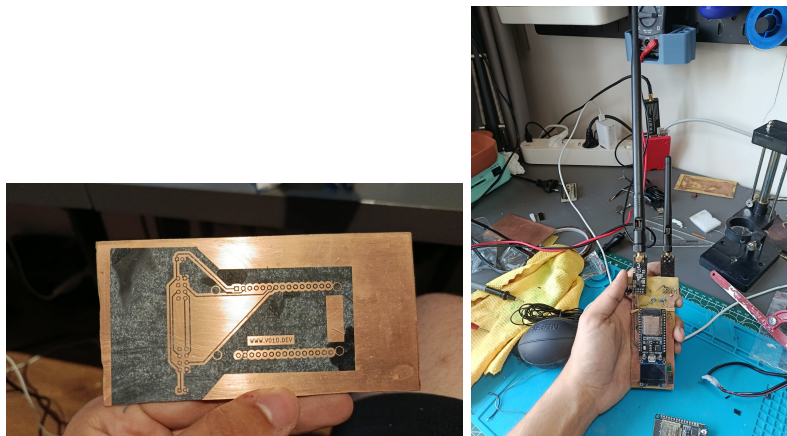


Figure 2: The first revision: toner transferred onto copper, and the same board assembled.

Flight Telemetry & Attitude Indicator

2026

An ESP32 telemetry system for small aircraft, pairing an **MS5611** barometer with an **MPU6050** inertial measurement unit. I fused the two sensors through a complementary filter (to correct the gyroscope's drift against the accelerometer's noise) and streamed attitude and altitude over serial at roughly 50 Hz.

The ground station is written in Python with **PySide6** and **Qt QML**, drawing a live artificial horizon from the incoming telemetry. I started it in Pygame and migrated it to QML partway through, because the original approach could not render smoothly enough to be useful as an instrument.

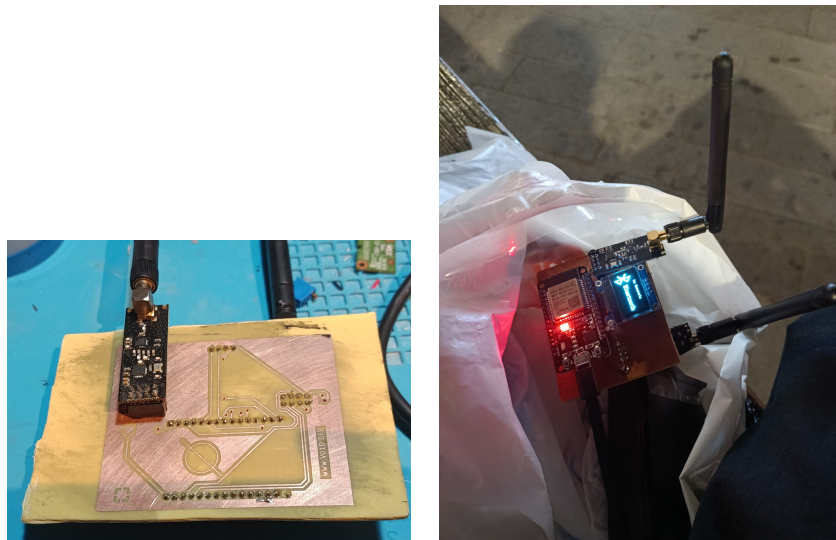


Figure 3: The final revision after etching, and the finished board in use.

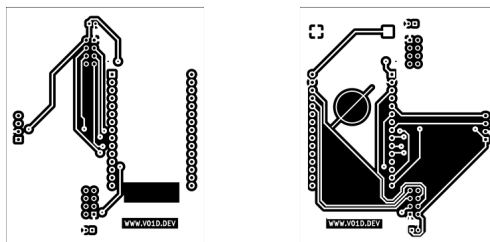


Figure 4: Front and back copper layers exported from KiCad. These are the etch artwork.



Figure 5: The telemetry board.

Discrete Transistor Logic Gates

2026

A board that implements **AND**, **NOT**, **OR** and **NAND** gates from individual transistors, with no logic ICs anywhere: seven **PN2222A** NPN transistors in resistor-transistor logic, with LEDs as outputs and tactile switches as inputs.

I built this to work below the abstraction layer. It is easy to use a 74-series chip and never think about what is inside it. Designing the same gates from transistors means reasoning about base current, saturation and voltage levels directly. I designed the schematic, routed the two-layer board with a

ground plane, and fabricated the first revision. It is currently under debug, and I am working through the fault.

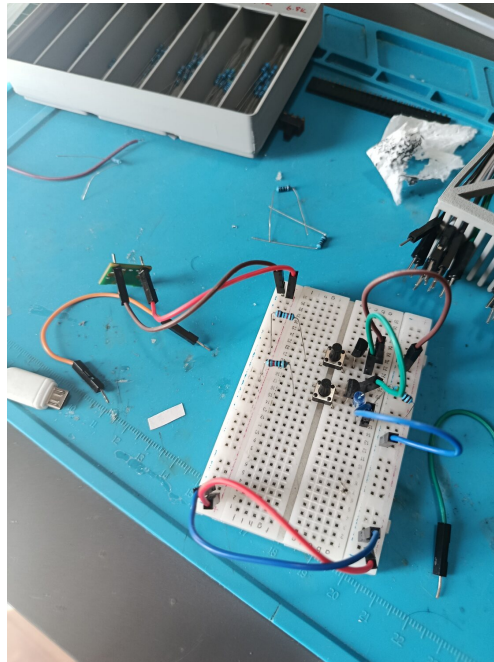


Figure 6: The gate circuits on a breadboard, before committing to the PCB.

RC Aircraft & Radio Link

2025–2026

Two fixed-wing designs, drawn and built from first principles rather than from a kit. **FCF-1** uses a 9 mm carbon fibre main spar with a **NACA 63412** airfoil, and I modelled the airframe in **FreeCAD**: wing structure, servo mounts, pushrods, control horns, aileron hinges and motor mount, all as parts that could actually be cut and printed. It has flown. A later design, **Simple-Glider**, was built to be easy to reproduce from foam board and printed parts, with a full build guide covering dimensions and part counts.

For the radio link on these aircraft I wrote the transmitter and receiver firmware myself: four analog channels encoded over **ESP-NOW** at 20 Hz, driving servos on the receiving end, with a failsafe that recentres every channel if the link is lost for three seconds.

Published KiCad Library

2026

The SSD1306 0.96-inch OLED module is one of the most common parts in hobby electronics, and I could not find a KiCad symbol and footprint for it that worked in a current version of KiCad. So I made one and published it: a native **KiCad 9** symbol and footprint, converted from an earlier library with the original **CC BY-NC-SA** attribution preserved as its licence requires.

It is not a file sitting in a drawer. I use this footprint in my own boards, including the RF board above.

Home PCB Fabrication

ongoing

Everything with a circuit board in this portfolio was made by hand, and the process itself has been a project. I generate artwork from KiCad, transfer it to copper, etch, drill and assemble, and I designed for that process on the boards above. I modelled a **toner transfer tool** in FreeCAD specifically for this workflow, and I write up the method on my site because the information that helped me most came from other people doing the same thing.

I also model and print mechanical parts, from enclosures for the boards above to a reversible USB power modification for a FlySky i6X transmitter that fits into the battery compartment without altering the case.

I took a Xiaomi router apart to understand how it boots. I located the UART header on the board and connected to it to capture the boot log, which is the most direct way to see what a device does before any of its own software takes over. The console turned out to be locked, so I moved to the flash instead and extracted the firmware image, which showed the firmware was built on an open-source base. I stopped there when the university term started.

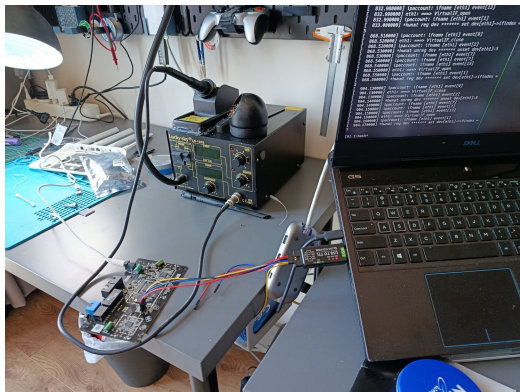


Figure 7: Capturing the boot log over UART.

Software Work

2025–2026

My professional experience is in software, and it is where I learned what production actually means.

At **Applab** in Doha I was the primary author of the backend for **Al Jazeera's** AI video editing and translation platform: a Quart API, an asynchronous job worker, and the queueing between them on Azure Container Apps, Cosmos DB and Storage Queues. I integrated Azure Cognitive Services and Azure OpenAI for transcription and translation across 14 locales, and built the failure handling (poison-message routing, bounded retry, per-action logging) that a long-running pipeline needs. Separately I built an AI chatbot for the **General Authority of Customs** on FastAPI, Semantic Kernel and Azure AI Search.

Outside work, I built a retrieval-augmented chatbot over university documentation using LangChain and MongoDB Atlas Vector Search, with an incremental ingest pipeline that hashes each page so only what changed gets reprocessed. I also maintain **TIPOT**, a PDF organising tool that has received and merged a pull request from an outside contributor.

Toolchain. C and C++ on ESP32/ESP8266, Python. KiCad 10 for schematic capture and PCB layout, including custom footprint authoring. FreeCAD for mechanical design, FDM printing for parts. SPI, I²C, UART, PWM and ADC at the hardware level; nRF24L01, ESP-NOW, ExpressLRS, Bluetooth and IR for radio. Bench work with a soldering iron, multimeter and serial telemetry.